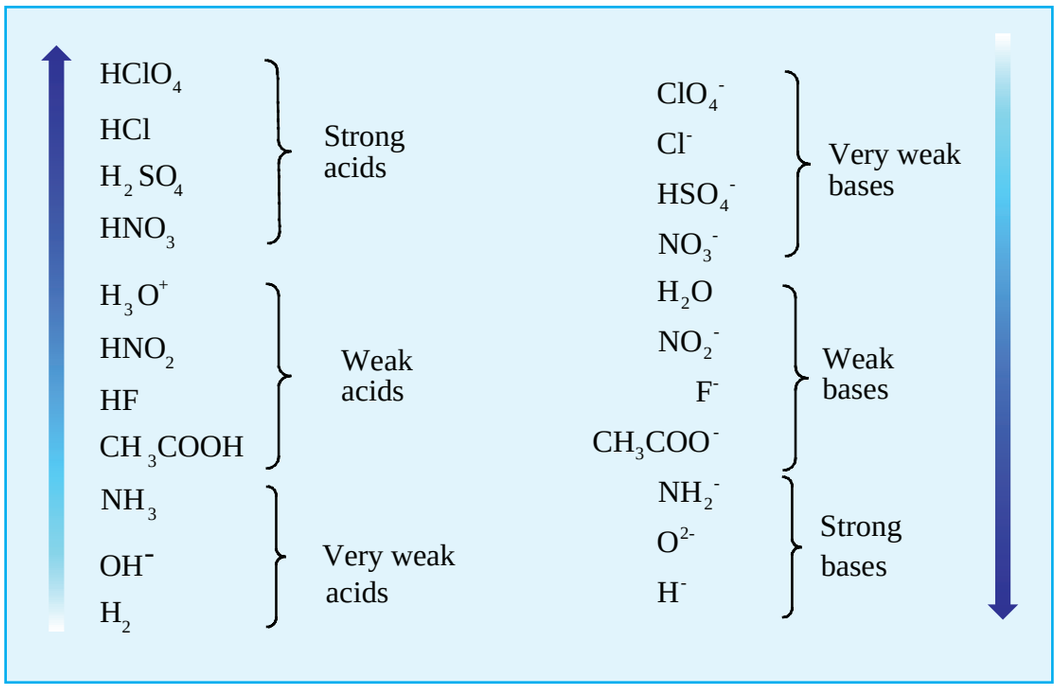
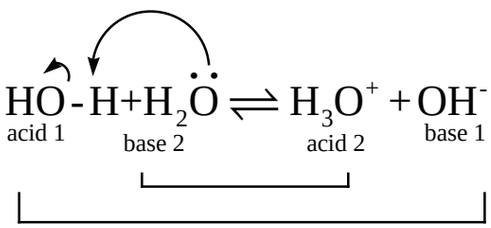


8.3 Ionisation of water

We have learnt that when an acidic or a basic substance is dissolved in water, depending upon its nature, it can either donate (or) accept a proton. In addition to that the pure water itself has a little tendency to dissociate. i.e, one water molecule donates a proton to an another water molecule. This is known as auto ionisation of water and it is represented as below.



Conjugate acid - base pairs

In the above ionisation, one water molecule acts as an acid while the another water molecule acts as a base.

The dissociation constant for the above ionisation is given by the following expression

$$K=\frac{[H_3O^+][OH^-]}{[H_2O]^2} \qquad .....(8.3)$$

The concentration of pure liquid water is one. i.e, $[H_2O]^2 = 1$

$$\because\ K_w=[H_3O^+][OH^-] \qquad .....(8.4)$$

Here, K_w represents the ionic product (ionic product constant) of water

It was experimentally found that the concentration of H_3O^+ in pure water is 1×10^{-7} at 25°C. Since the dissociation of water produces equal number of H_3O^+ and OH^-, the concentration of OH^- is also equal to 1×10^{-7} at 25°C.

XII U8-Ionic equilibrium.indd 7 11-01-2021 12:58:20